UNIFORM SMD BOUND FOR COHERENT SHEAVES ON SMOOTH PROPER SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Borrows the inequality proven inside the proof of the strong generator card (same batch), restated explicitly in-span.

Allowed standard prerequisites (editorial).

- Strong generator. lemma-smooth-proper-strong-generator L26462 (carded this batch). Source of the restated inequality.

Source reference. Original reference 0FX0: focused-seeds/geometry/S9-eqv-diagonal-trick-proper/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZ6: sources/stacks-equiv/pinned-original-source.tex, source lines 1995–1995. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X be a proper smooth scheme over k. There exists integers $m, n \geq 1$ and a finite locally free $\mathcal{O}_X$-module $\mathcal{G}$ such that every coherent $\mathcal{O}_X$-module is contained in $smd(add(\mathcal{G}[-m, m])^{\star n})$ with notation as in Derived Categories, Section 0FX0.*

Proof. In the proof of Lemma 0FZ6 we have shown that there exist $m', n \geq 1$ such that for any coherent $\mathcal{O}_X$-module $\mathcal{F}$,

$$\mathcal{F} \in smd(add(\mathcal{G}[-m' + a, m' + b])^{\star n})$$

for any $a \leq b$ such that $H^i(X, \mathcal{F})$ is nonzero only for $i \in [a, b]$. Thus we can take $a = 0$ and $b = \dim(X)$. Taking $m = \max(m', m' + b)$ finishes the proof. □

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.